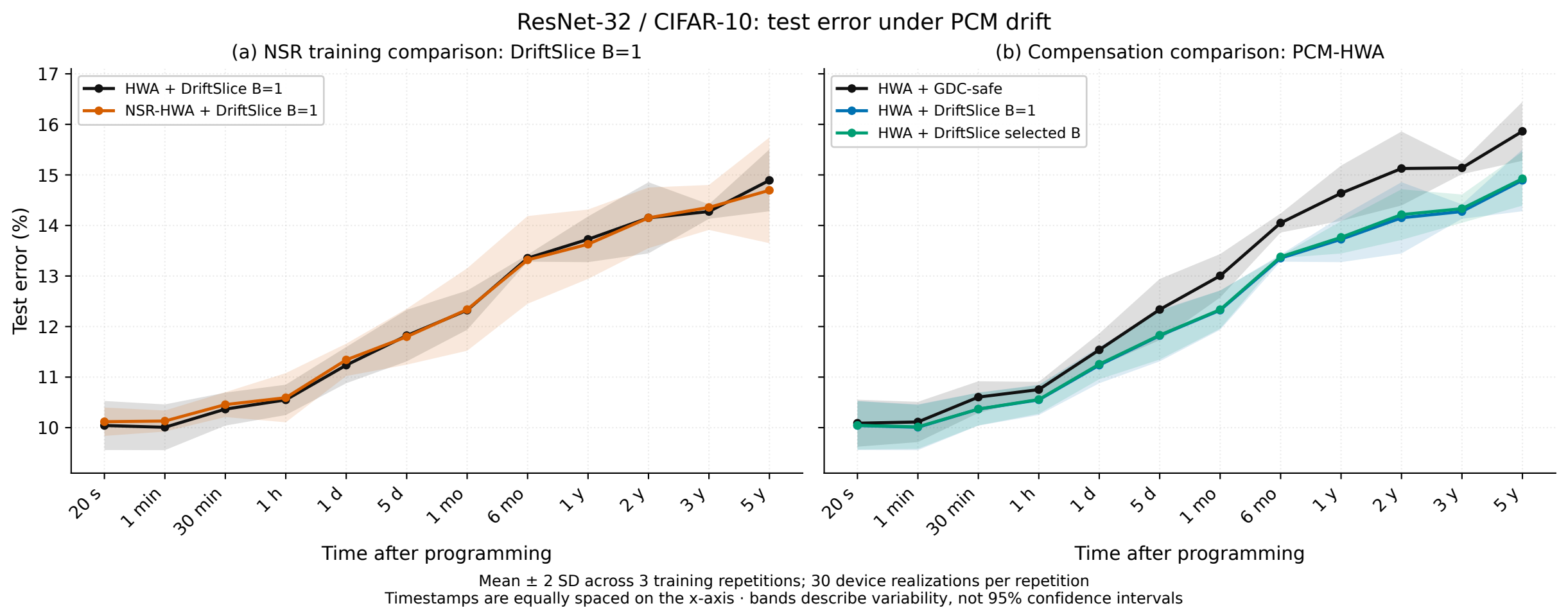
ResNet-32 / CIFAR-10: test error under PCM drift
(a) NSR training comparison: DriftSlice B=1
(b) Compensation comparison: PCM-HWA
Test error (%)
17
16
15
14
13
12
11
10
HWA + DriftSlice B=1
NSR-HWA + DriftSlice B=1
HWA + GDC-safe
HWA + DriftSlice B=1
HWA + DriftSlice selected B
20 s
1 min
30 min
1 h
1 d
5 d
1 mo
6 mo
1 y
2 y
3 y
5 y
Time after programming
Time after programming
Mean ± 2 SD across 3 training repetitions; 30 device realizations per repetition
Timestamps are equally spaced on the x-axis · bands describe variability, not 95% confidence intervals